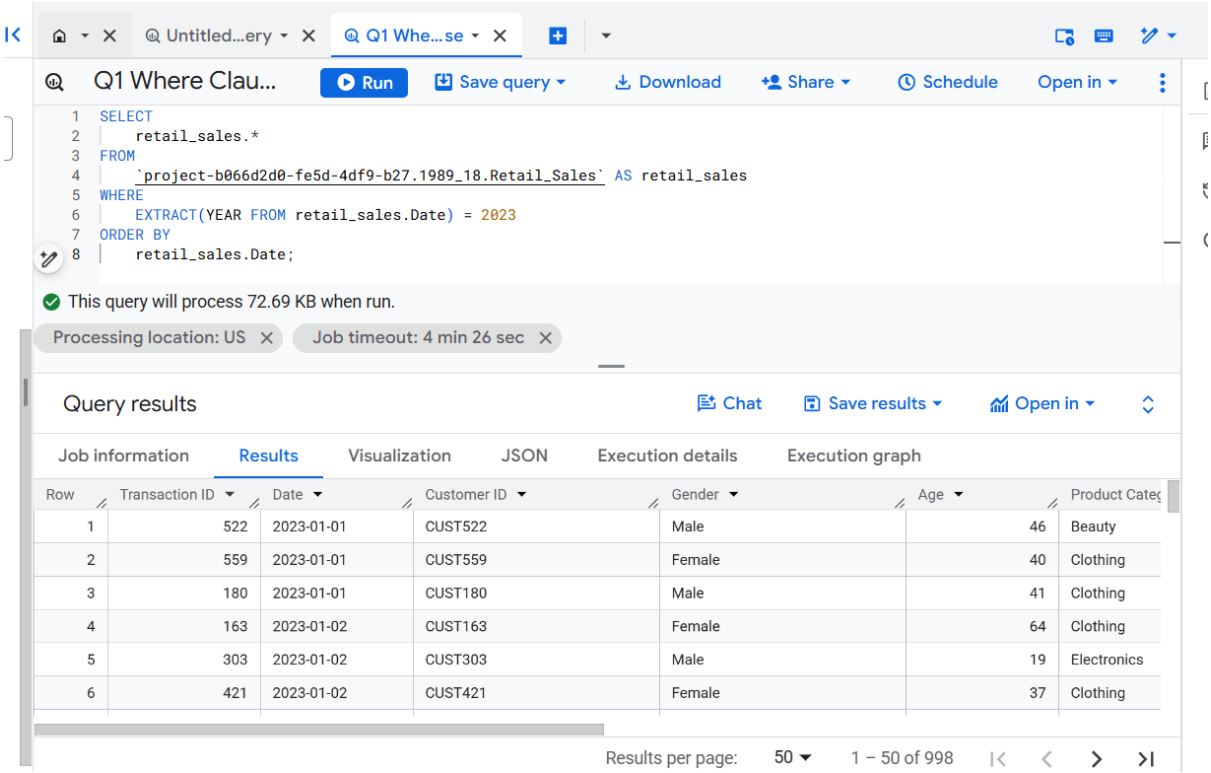
Practical Exercise

Google BigQuery

Question 1 – Where Clause

Filter all transactions that occurred in the year 2023.



Question 2 - Filtering + Conditions

Display all transactions where the Total Amount is more than the average Total Amount of the entire dataset.

Untitled query

```sql
SELECT
    sales.`Transaction ID`,
    sales.Date,
    sales.`Customer ID`,
    sales.Gender,
    sales.Age,
    sales.`Product Category`,
    sales.Quantity,
    sales.`Price per Unit`,
    sales.`Total Amount`
FROM
    `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS sales
WHERE
    sales.`Total Amount` > (
        SELECT
            AVG(sub.`Total Amount`)
        FROM
            `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS sub
    )
ORDER BY
    sales.Date;
```

This query will process 72.69 KB when run.

Using on-demand processing quota | Processing location: US | Job timeout: 4 min 27 sec

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Results per page: 50 | 1 – 50 of 350

Question 3 - Aggregate Functions

Calculate the total revenue (sum of Total Amount).

Question 3

```sql
SELECT
    SUM(sales.`Total Amount`) AS Total_Revenue
  FROM
    `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS sales
```

This query will process 7.81 KB when run.

Using on-demand processing quota | Processing location: US | Job timeout: 4 min 29 sec

Query results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Total_Revenue
1	456000

Question 4 – DISTINCT

Display all distinct Product Categories in the dataset.

Question 4 | Run | Save query | Download | Share | Schedule | Open in | More

```sql
SELECT DISTINCT
    `Retail_Sales`.`Product Category` AS Product_Category
FROM
    `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS Retail_Sales;
```

This query will process 10.17 KB when run.

Using on-demand processing quota | Processing location: US | Job timeout: 4 min 28 sec

Query results | Chat | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product_Category
1	Beauty
2	Clothing
3	Electronics

Question 5 - GROUP BY

For each Product Category, calculate the total quantity sold.

Question 5 | Run | Save query | Download | Share | Schedule | Open in

```google_sql
SELECT
  Retail_Sales.`Product Category` AS Product_Category,
  uantity
FR
  `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS Retail_Sales
GROUP BY
  Product_Category;
```

Open resource side-by-side. (ctrl + click)

This query will process 17.98 KB when run.

Using on-demand processing quota | Processing location: US | Job timeout: 4 min 28 sec

Query results | Chat | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product_Category	Total_Quantity
1	Beauty	771
2	Clothing	894
3	Electronics	849

Results per page: 50 | 1 – 3 of 3

Question 6 - CASE Statement

Create a column called Age_Group that classifies customers as:

– 'Youth' (<30)

– 'Adult' (30–59)

– 'Senior' (60+)

Question... Run Save query Download Share Schedule Open in More

```sql
SELECT DISTINCT
    sales.`Customer ID`,
    sales.Age,
    CASE
        WHEN sales.Age < 30 THEN 'Youth'
        WHEN sales.Age BETWEEN 30 AND 59 THEN 'Adult'
        WHEN sales.Age >= 60 THEN 'Senior'
    END AS Age_Group
FROM
    `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS sales
```

This query will process 16.6 KB when run.

Using on-demand processing quota | Processing location: US | Job timeout: 4 min 27 sec

Query results Chat Save results Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Age_Group
1	CUST191	64	Senior
2	CUST204	39	Adult
3	CUST230	54	Adult
4	CUST232	43	Adult
5	CUST309	26	Youth

Results per page: 50 1 – 50 of 1000

Question 7 - Conditional Aggregation

For each Gender, count how many high-value transactions occurred (where Total Amount > 500).

Question 7

```
SELECT
    Retail_Sales.Gender,
    COUNTIF(Retail_Sales.`Total Amount` > 500) AS High_Value_Transactions
FROM
    `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS Retail_Sales
GROUP BY
    Retail_Sales.Gender;
```

This query will process 14.67 KB when run.

Using on-demand processing quota | Processing location: US | Job timeout: 4 min 23 sec

Query results

Row	Gender	High_Value_Transactions
1	Male	144
2	Female	155

Question 8 - HAVING Clause

For each Product Category, show only those categories where the total revenue exceeds 5,000

Untitled query

```
SELECT
    Retail_Sales.`Product Category` AS Product_Category,
    SUM(Retail_Sales.`Total Amount`) AS Total_Revenue
FROM
    `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS Retail_Sales
GROUP BY
    Product_Category
HAVING
    Total_Revenue > 5000
ORDER BY
    Total_Revenue DESC;
```

This query will process 17.98 KB when run.

Using on-demand processing quota | Processing location: US | Job timeout: 4 min 25 sec

Query results

Row	Product_Category	Total_Revenue
1	Electronics	156905
2	Clothing	155580
3	Beauty	143515

Results per page: 50 1 – 3 of 3

Question 9 - Calculated Fields

Display a new column called Unit_Cost_Category that labels a transaction as:

– 'Cheap' if Price per Unit < 50

– 'Moderate' if Price per Unit between 50 and 200

– 'Expensive' if Price per Unit > 200

Untitled...ery | *Question 9

Question 9 | Run | Save query | Download | Share | Schedule | Open in

```sql
SELECT
    Retail_Sales.`Transaction ID` AS Transaction_ID,
    Retail_Sales.`Price per Unit` AS Price_per_Unit,
    CASE
        WHEN Retail_Sales.`Price per Unit` < 50 THEN 'Cheap'
        WHEN Retail_Sales.`Price per Unit` BETWEEN 50 AND 200 THEN 'Moderate'
        WHEN Retail_Sales.`Price per Unit` > 200 THEN 'Expensive'
    END AS Unit_Cost_Category
FROM
    `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS Retail_Sales;
```

This query will process 15.63 KB when run.

Using on-demand processing quota | Processing location: US | Job timeout: 4 min 28 sec

Query results | Chat | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction_ID	Price_per_Unit	Unit_Cost_Category
1	191	25	Cheap
2	204	25	Cheap
3	230	25	Cheap
4	232	25	Cheap
5	309	25	Cheap
6	310	25	Cheap

Results per page: 50 | 1 – 50 of 1000

Question 10 - Combining WHERE + CASE

Display all transactions from customers aged 40 or older and add a column Spending_Level showing:

– 'High' if Total Amount > 1000

– 'Low' otherwise

Untitled...ery | *Question10

Question 10 | Run | Save query | Download | Share | Schedule | Open in | More

```
SELECT
    retail_sales.`Customer ID`,
    retail_sales.`Age`,
    retail_sales.`Total Amount`,
    CASE
        WHEN retail_sales.`Total Amount` > 1000 THEN 'High'
        ELSE 'Low'
    END AS Spending_Level
FROM
    `project-b066d2d0-fe5d-4df9-b27.1989_18.Retail_Sales` AS retail_sales
WHERE
    retail_sales.`Age` >= 40;
```

This query will process 24.42 KB when run.

Using on-demand processing quota | Processing location: US | Job timeout: 4 min 27 sec

Query results | Chat | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Total Amount	Spending_Level
1	CUST191	64	25	Low
2	CUST230	54	25	Low
3	CUST232	43	25	Low
4	CUST363	64	25	Low

Results per page: 50 | 1 – 50 of 558